

Small and Simple Molecular Structure Based Thermally Stable Ruthenium Precursor in Advancing Ruthenium ALD Process for Scaled Interconnect Metallization

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1 Introduction

In the last few decades, continued down-scaling of integrated circuits (ICs) following Moore’s law has significantly increased interconnect density. This drives the need for advanced interconnect miniaturization and requires novel materials to mitigate parasitic effects and RC delays on semiconductor devices at the nanoscale. Copper (Cu) has been widely used for wiring in the semiconductor field since the late 1990s. The challenging long electron mean free path for Cu, compared to ruthenium (Ru), causes severe deterioration of the semiconductor device functionality when the dimension is scaled down to below a few nanometers. With the demand for more Moore and continuous downscaling of ICs, the channel material must be shrunk under the channel length, limiting Cu metallization. This is primarily due to the drastic increase of Cu’s resistivity arising from the scattered electrons at the surface and grain boundaries when the semiconductor device dimensions are smaller than Cu’s electron mean free paths, i.e., ≈ 40 nm. Moreover, Cu itself is susceptible to other reliability concerns,

including electromigration (EM), and time-dependent dielectric breakdown, due to its inherent chemical and thermal characteristics. These issues are aggravated by continued scaling via wiring miniaturizations. Therefore, to overcome these challenges, introducing a small yet simple molecular structure based novel Ru precursor without involving any oxygen in the structure with high thermal stability is imperative to withstand the high-temperature ALD-Ru process. This precursor achieves a high GPC (≈ 1.28 Å/cycle) and a short incubation period

(8 cycles) with thermal stability up to 400 °C. Crystallographic analysis via electron backscatter diffraction (EBSD), next-generation aberration-corrected ultrahigh-resolution transmission electron microscope (UHR-TEM), secondary ion mass spectrometry, and advanced X-ray diffraction and modeling based on the Fuchs-Sondheimer and Mayadas-Shatzkes (FS-MS) approach revealed that the grain growth and the formation of low-energy grain boundaries, such as

coincidence site lattice (CSL) boundaries, are critical to resistivity reduction approaching bulk Ru resistivity values. These microstructural improvised features are effectively promoted by high-temperature ALD with excellent substrate selectivity, made possible by the high thermal stability of the new precursor. This unique combination of high growth rate, low resistivity, thermal robustness, and substrate selectivity positions this novel Ru precursor as a highly promising candidate for a high-performance ALD-Ru process for next-generation interconnect semiconductor technology.

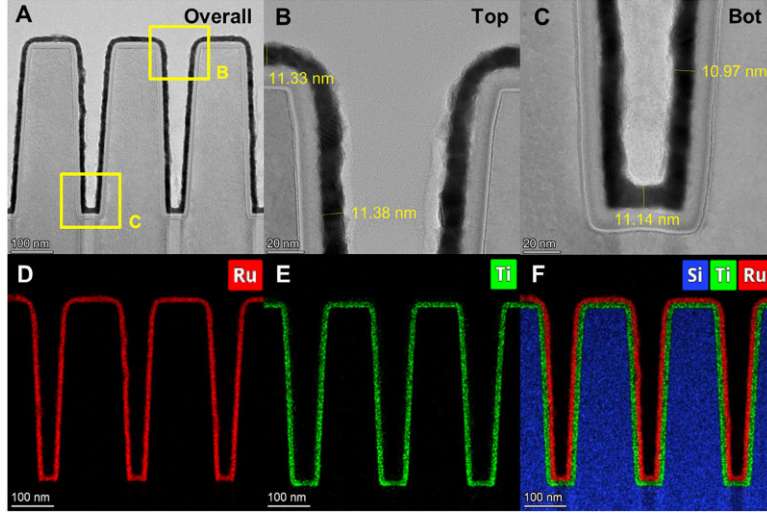


Figure 1: Ru interconnect

2 Conclusion

In summary, we have established a novel small and simple molecular ligand structure Ru precursor that overcomes key limitations in ALD-Ru process design for next-generation interconnect applications. This precursor demonstrates exceptional thermal stability up to 400 °C, enabling high-temperature ALD-Ru processes that enhance film densification, grain growth, and electrical performance. With a high growth per cycle (≈ 1.28 Å/cycle) and short incubation period (8 cycles) on TiN, the process achieves low resistivity values approaching bulk Ru down to 10.6 without annealing, and 8.65 after postdeposition hydrogen annealing. The newly synthesized Ru precursor also exhibits excellent substrate selectivity, with no Ru nucleation on SiO₂ even after 1000 ALD cycles at 300 °C, supporting area-selective integration strategies into the complex device architectures. These improvements are attributed to enhanced crystallinity, grain growth, and the formation of low-energy coincidence site lattice grain boundaries, which reduce electron scattering revealed by microstructural

analysis via EBSD, advanced aberration-corrected UHR-TEM/STEM, FS-MS modeling, and MLIP-based energy profile calculations. Compared to previously reported Ru-ALD processes, which often suffer from low growth rates, long incubation periods, and high film resistivity due to poor thermal stability, this new Ru precursor offers a balanced solution toward integrating low resistance, barrierless metal interconnects in advanced semiconductor nodes. [1]

References

- [1] Soo-Hyun Kim. Small and simple molecular structure based thermally stable ruthenium precursor in advancing ruthenium ald process for scaled interconnect metallization. *Advanced Science*, 2025.